Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 6

Title: Sorting & Filtering Data

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Sorting & Filtering Data

GitHub Repository Link: *https://github.com/anuskaGHS/Business-Analytics*

Problem Statement

A sales manager wants to identify the company's highest-value transactions and underperforming products. Use sorting and filtering techniques to extract relevant business information.

Dataset used: sales_analysis_dataset.xlsx

Methodology / Steps Followed

Step 1: Data Collection and Loading

The provided Excel workbook (sales_analysis_dataset.xlsx) was loaded into Power BI Desktop using Get Data → Excel workbook. Only the Sales_Data sheet (raw transactions) was selected. It contains 300 transactions, 14 columns, 12 products and 4 regions for FY Oct 2025 – Sep 2026: Transaction ID, Date, Region, Salesperson, Customer, Product, Category, Quantity, Unit Price, Discount, Unit Cost, Revenue, Profit, Returned.

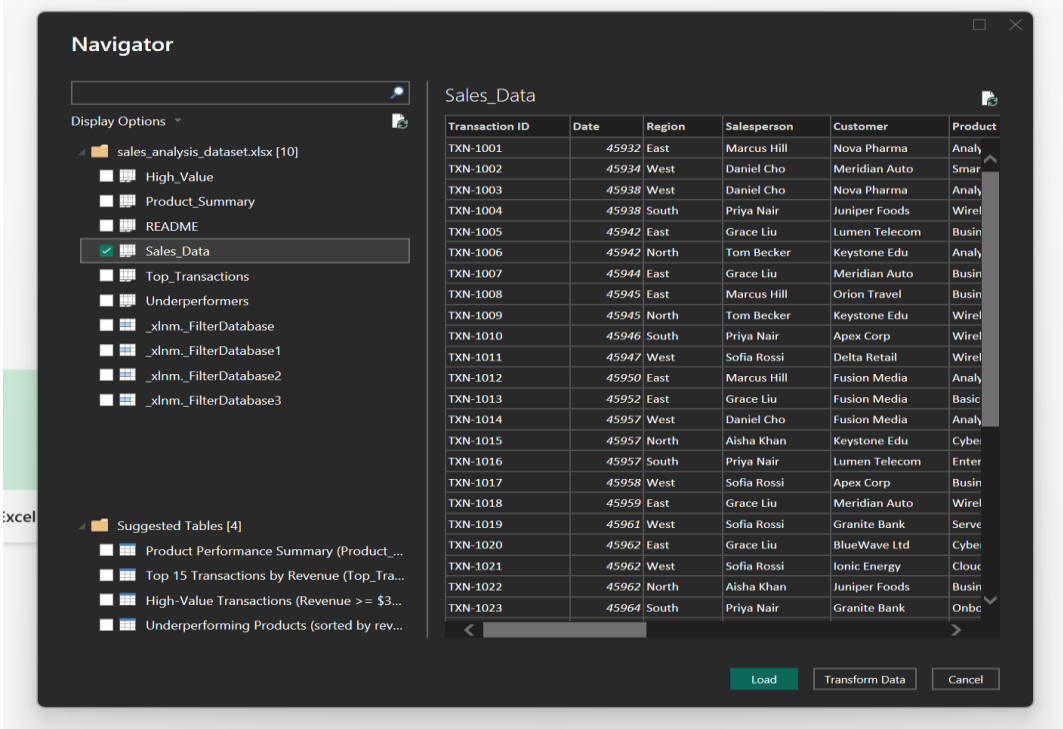


Figure 1: Data Load (Navigator window)

Step 2: Data Preparation in Power Query Editor

- The table was renamed to Sales.
- **Columns were renamed:** Unit Price ($) → Unit Price, Unit Cost ($) → Unit Cost, Revenue ($) → Revenue, Profit ($) → Profit, Discount % → Discount.
- **Data types:** Date → Date; Unit Price, Unit Cost, Revenue, Profit → Fixed Decimal Number; Discount → Percentage; Returned kept as Text (Y/N).
- Column quality showed 100% valid values, with no errors and no empty values. Close & Apply was then clicked.

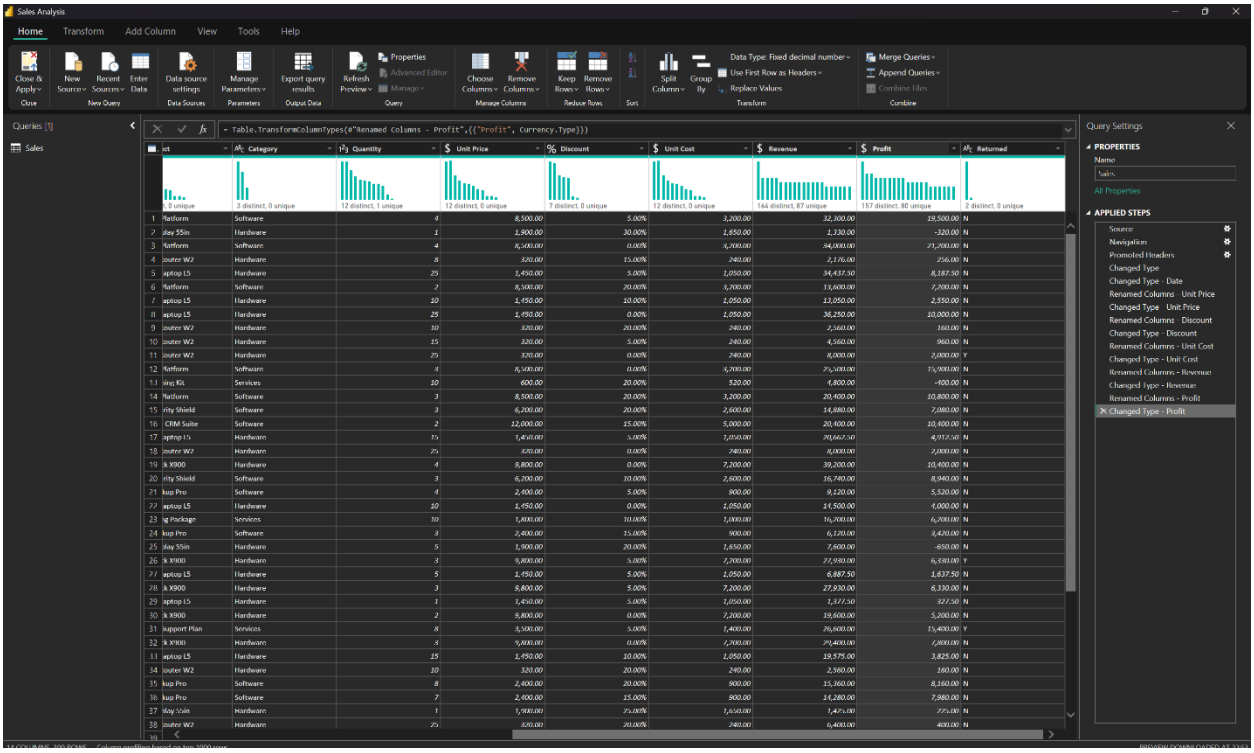


Figure 2: Power Query Editor

Step 3: DAX Measures

1. **Total Revenue** = SUM(Sales[Revenue])
2. **Total Profit** = SUM(Sales[Profit])
3. **Profit Margin %** = DIVIDE([Total Profit], [Total Revenue])
4. **Total Transactions** = DISTINCTCOUNT(Sales[Transaction ID])
5. **High-Value Threshold** = 32400

6. **High-Value Transactions =**

   VAR t = [High-Value Threshold]

   RETURN CALCULATE([Total Transactions], Sales[Revenue] >= t)

7. **High-Value Revenue =**

   VAR t = [High-Value Threshold]

   RETURN CALCULATE([Total Revenue], Sales[Revenue] >= t)

8. **High-Value Revenue Share % =** DIVIDE([High-Value Revenue], [Total Revenue])
9. **Avg Product Revenue =** CALCULATE(AVERAGEX(VALUES(Sales[Product]), [Total Revenue]), ALL(Sales))
10. **Underperforming Flag =**

    IF(HASONEVALUE(Sales[Product]),

    IF([Total Revenue] < 0.6 * [Avg Product Revenue] || [Profit Margin %] < 0.15, 1, 0))

A high-value transaction is defined as revenue of $32,400 or more (about the top 10% of transactions). An underperforming product has revenue below 60% of the average product revenue, or a profit margin below 15%.

Step 4: Sorting and Filtering Techniques Applied

- **Top N filter:** Transaction ID filtered to the Top 10 by Total Revenue.
- **Sorting:**
  - transactions sorted in descending order of revenue (highest first);
  - underperforming products sorted in ascending order of revenue (lowest first);
  - products sorted by profit in the profit chart.
- **Visual-level (measure-based) filter:** Underperforming Flag = 1.
- **Slicers:** Region, Category and Salesperson, for interactive filtering.

Step 5: Dashboard Development

Page 1 – High-Value Transactions:

- KPI cards (Total Revenue, Total Profit, Profit Margin %, High-Value Transactions,

High-Value Revenue Share %)

- Top 10 Transactions table
- High-Value Revenue by Product (bar chart)
- High-Value Transactions by Region (column chart)
- Slicers

Page 2 – Underperforming Products:

- Table of underperforming products and
- Profit by Product column chart

Output

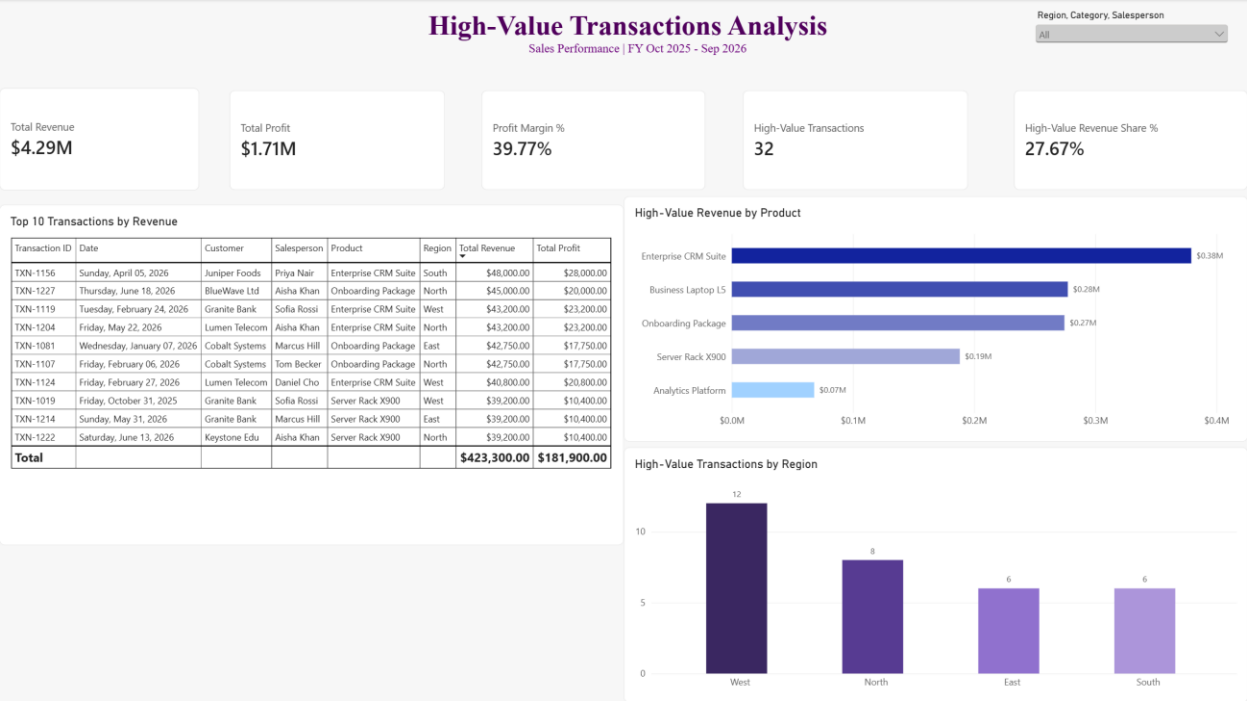


Figure 3: Dashboard Page 1 – High-Value Transactions

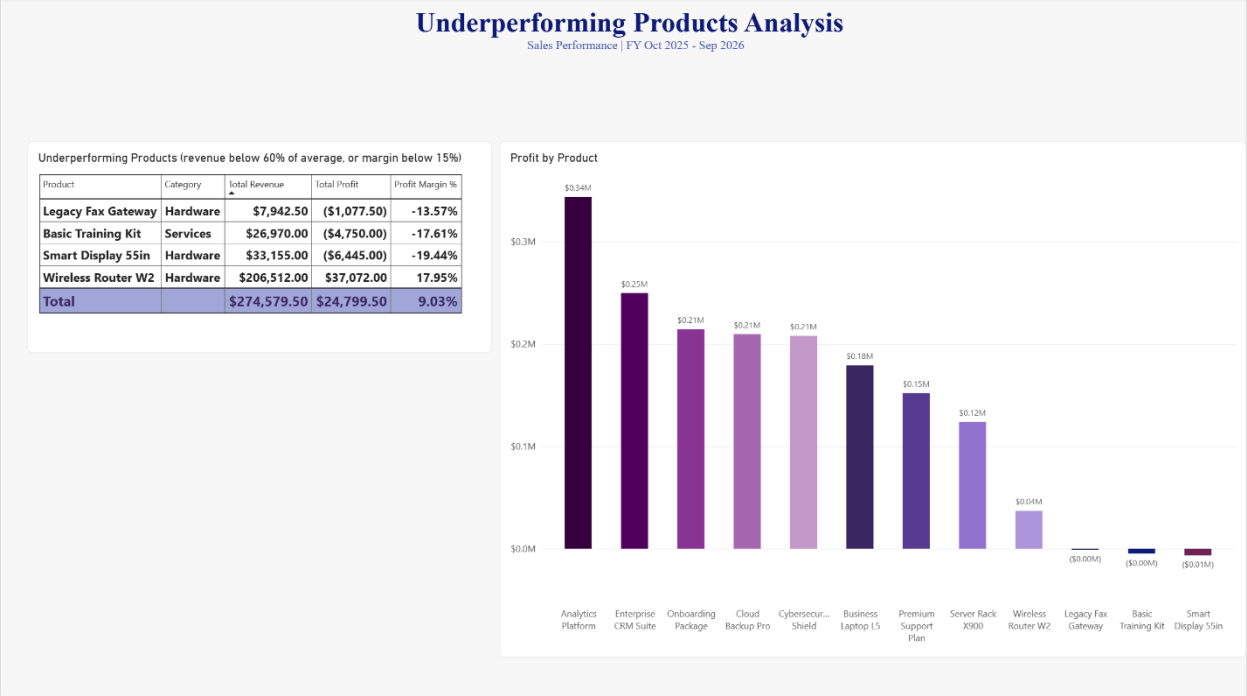


Figure 4: Dashboard Page 2 – Underperforming Products

Business Insights

Overall: Total revenue is $4.29M and total profit is $1.71M, a profit margin of 39.77%.

Highest-value transactions:

- The top transaction is TXN-1156 for $48,000 (Enterprise CRM Suite, South region, Priya Nair).
- The Top 10 transactions together bring in $423,300 (about 9.9% of total revenue).
- 32 transactions (about 10.7% of all 300) are high-value (≥ $32,400), yet they contribute $1.19M, which is 27.67% of total revenue.
- By product, Enterprise CRM Suite leads high-value revenue (about $0.38M), followed by Business Laptop L5 ($0.28M), Onboarding Package ($0.27M), Server Rack X900 ($0.19M) and Analytics Platform ($0.07M).
- By region, West has the most high-value transactions (12), then North (8), East (6) and South (6).

Underperforming products:

- Four products are flagged: Legacy Fax Gateway ($7,942.50 revenue), Basic Training Kit ($26,970), Smart Display 55in ($33,155) and Wireless Router W2 ($206,512).

- The first three are loss-making (negative profit), with a combined loss of $12,272.50. Their margins are -13.57%, -17.61% and -19.44%.
- Wireless Router W2 is profitable (17.95% margin) but is flagged because its revenue is below 60% of the average product revenue ($357,493.50).
- Together the four products contribute only about 6.4% of total revenue ($274,579.50).

Conclusion

Using sorting, Top N filtering, measure-based filters and slicers in Power BI, the highest-value transactions and the underperforming products were identified. A small share of transactions (32 of 300) generates over a quarter of revenue, so protecting those deals is important. Four products are underperforming, three of which make losses, so the manager can review their pricing, discounts and continuation. The dashboard lets the manager filter by region, category and salesperson.